

Chapter 2: Communication and networking technologies

Learning objectives

By the end of this chapter you should be able to:

- show understanding of the purpose and benefits of networking devices
- show understanding of the characteristics of a LAN (local area network) and a WAN (wide area network)
- explain the client-server and peer-to-peer models of networked computers
- show understanding of thin-client and thick-client and the differences between them
- show understanding of the bus, star, mesh and hybrid topologies
- show understanding of cloud computing
- show understanding of the differences between and implications of the use of wireless and wired networks
- describe the hardware that is used to support a LAN
- describe the role and function of a router in a network
- show understanding of Ethernet and how collisions are detected and avoided
- show understanding of bit streaming
- show understanding of the differences between the World Wide Web (WWW) and the Internet
- describe the hardware that is used to support the Internet
- explain the use of IP addresses in the transmission of data over the Internet
- explain how a Uniform Resource Locator (URL) is used to locate a resource on the World Wide Web (WWW) and the role of the Domain Name Service (DNS).

2.01 The evolution of the purpose and benefits of networking

Wide area network (WAN)

During the 1970s it would be normal for a large organisation to have a computer. This computer would be a mainframe or minicomputer. The computer could have been running a time-sharing operating system with individual users accessing the computer using a terminal connected to the computer with a cable. Technology was developed that allowed computers in different organisations to be networked using what would now be described as a **wide area network (WAN)**. In a WAN, the networked computers could be thousands of kilometres apart.

The benefits of having the computers connected by a WAN were:

- a 'job' could be run on a remote computer that had the required application software
- a data archive that was stored on a remote computer could be accessed
- a message could be transmitted electronically to a user on a remote computer.

Today, a typical WAN is characterised by the following.

- It will be used by an organisation or a company to connect sites or branches.
- It will not be owned by the organisation or company.
- It will be leased from a public switched telephone network company (PSTN).
- A dedicated communication link will be provided by the PSTN.
- The transmission medium will be fibre-optic cable.
- Transmission within the WAN will be from switch to switch.
- A switch will connect the WAN to each site.
- There will not be any end-systems connected directly to the WAN.

Local area network (LAN)

In the 1980s the arrival of the microcomputer or personal computer (PC) changed computing. In an organisation, a user could have their own computer on their desk. The computer could be a stand-alone system or some organisations chose to have more than one computer connected using a **local area network (LAN)**. It was called a local area network because it typically connected PCs that were in one room or in one building or on one site.



KEY TERMS

Wide area network (WAN): a network connecting computers on different sites, possibly thousands of kilometres apart

Local area network (LAN): a network connecting computers in a single room, in a single building or on a single site

The benefits of connecting PCs in a LAN included the following.

- The expense of installing application software on each individual PC could be saved by installing the software on an application server attached to the LAN instead.
- A file server could be attached to the LAN that allowed users to store larger files and also allowed files to be shared between users.
- Instead of supplying individual printers to be connected to a user's PC, one or more printers could be attached to a print server that was connected to the LAN; these could be higher quality printers.

- Managers in organisations could use electronic mail to communicate with staff rather than sending round memos on paper.
- The 'paper-less office' became a possibility, where files were to be stored in digital form on a file server rather than as paper copies in a filing cabinet.

Today, a typical LAN is characterised by the following.

- It will be used by an organisation or a company within a site or branch.
- It will be owned by the organisation or company.
- It will be one of many individual LANS at one site.
- The transmission medium will be twisted pair cable or WiFi.
- The LAN will contain a device that allows connection to other networks.
- There will be end-systems connected which will be user systems or servers.

Discussion Point:

If a print server was attached to a network, what functionality could it provide?

Internet working

The 1990s can be said to be when the modern era of computing and network use started, with the beginning of widespread use of the Internet. The word Internet is a shortened form of the term 'internetwork', which describes a number of networks all connected together. LANs are connected to WANs which are in turn connected to the Internet to allow access to resources world-wide. The other technologies defining the modern era, namely mobile devices and wireless networking, started to become commonly used in the 2000s.

The purpose and benefits of networking have not changed but their scale and scope has increased enormously. In particular, people now have full access to networks from their personal devices.

The client-server model

The **client-server** model (or architecture) was first used in large organisations when they had installed internal networks. Typically, the organisation would have individual LANs connected via an organisation-wide WAN. An individual LAN might have had an application server attached. The organisation was likely to need a powerful central computer. The central computer could be connected to the WAN as a server. It would probably not have individual users connected to it directly. A PC, attached to a LAN, could access the server as a client.

The client-server mode of operation nowadays is different. The client is a web browser connected to the Internet. The server is a web server hosted on the Internet.

The server provides an application and the client uses the application. There are two options for how the client functions.

A **thin-client** is one which:

- chooses an application to run on the server
- sends input data to the server when requested by the application
- receives output from the application.

KEY TERMS

Client-server: an architecture where a client runs an application provided by a server on a network

Thin-client: a client that only provides input and receives output from the application

A **thick-client** is one which:

- chooses an application provided by the server
- possibly carries out some processing before running the application on the server and also after receiving output from the application
- alternatively, possibly downloads the application from the server and runs the application itself.



TIP

In thick-client mode the processing on the client can be controlled by the use of a scripting language. You do not need to know any details of this.



KEY TERM

Thick-client: a client that carries out at least some of the processing itself

The client-server approach is the choice in the following circumstances.

- The server stores a database which is accessed from the client system.
- The server stores a web application which allows the client system to find or, sometimes, supply information.
- The server stores a web application which allows the client system to carry out an e-commerce or financial transaction.

File sharing

If a user uploads files to a file server then the client-server operation can be used by another user to download these from the server.

An alternative mode of operation for sharing files is peer-to-peer networking. Instead of having one server that many clients access, a peer-to-peer network operates with each peer (networked computer) storing some of the files. Each peer can therefore act as a client and request a file from another peer or it can act as a server when another peer requests the download of a file.

The peer-to-peer model has several advantages compared to client-server file downloading:

- it avoids the possibility of congestion on the network when many clients are simultaneously attempting to download files
- parts of a file can be downloaded separately
- the parts are available from more than one host.

The client-server model has the following advantages.

- It allows an organisation to control the downloading and use of files.
- The files can be better protected from malware attacks because the files are stored on one server which will be regularly scanned using appropriate anti-virus software.

2.02 Network topologies

There are five requirements for a data communications system: a sender, a receiver, a transmission medium, a message and a protocol (see Chapter 17 for details about protocols). A transmission medium can be air (e.g. for WiFi) or cables (e.g. for Ethernet). Data can be sent through the medium in different modes:

- simplex mode where data flow is one-way only
- half duplex where data can flow either way but not simultaneously
- full duplex where simultaneous both-ways data flow is possible.

A 'message' is any type of data, which can be sent as either:

- a broadcast, which is a one-to-all communication (as used traditionally for radio and television)
- a multicast, which is from one source to many destinations
- a unicast, which is a one-to-one communication.

A data communications system may consist of a single isolated network. There are several possibilities for the **topology** of an isolated network. The simplest of these is where two systems are connected by a network link as shown in Figure 2.01. This is an example of a point-to-point connection, which is a dedicated link. Transmission might be simplex or duplex and a message can only be unicast.



Figure 2.01 A point-to-point network

Early LAN topologies used either a ring or a **bus topology**. We don't need to cover the ring topology as it is not used very often now. A bus topology has only one link but it is shared by a number of end-systems and is therefore described as a multi-point connection. The configuration is shown in Figure 2.02. There is no direct connection between any pair of end-systems. A message must therefore be broadcast even though it might only be intended for one **end-system**. The topology is resilient because a fault in an end-system or in the link to it does not affect the use of the network by the other end-systems.



Figure 2.02 A bus network

An example of a fully-connected **mesh topology** is shown in Figure 2.03. In this configuration, each end-system has a point-to-point connection to each of the other end-systems. Transmission is duplex; messages might be unicast, multicast or broadcast.

KEY TERMS

Topology: the configuration of a network that defines how the various devices on the network are connected

Bus topology: contains one shared link to which all devices are attached

End-system: a computer or server connected to a network

Mesh topology: contains direct links between devices

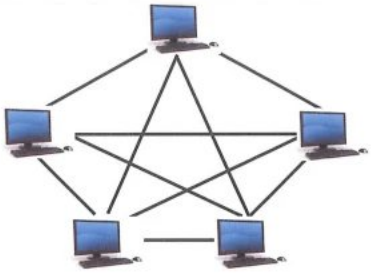


Figure 2.03 A mesh network.

Figure 2.03 shows end-systems connected in a mesh topology but this is unrealistic because of the amount of cabling required. A mesh topology can be used when individual LAN switches are connected in a network. The topology is essential for the connection of routers within the infrastructure of the Internet.

The final possibility is a **star topology** which is shown in Figure 2.04.

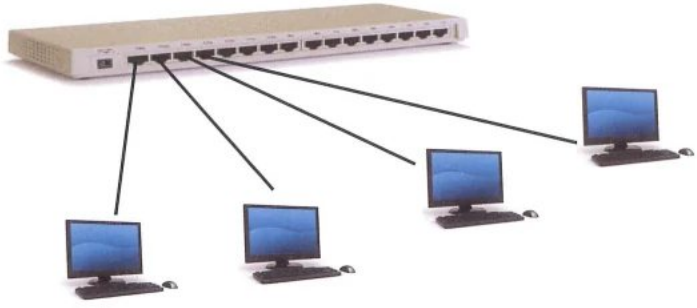


Figure 2.04 A star topology

KEY TERM

Star topology: each end-system is linked to a central device

Figure 2.04 could have been drawn so that it looked like a star but has been drawn to show the physical configuration that is used in a real life installation. In a star topology, each end-system has a point-to-point connection to the central device. Transmission is duplex and messages from the central device might be unicast, multicast or broadcast. As with the bus topology, the failure of an end-system, or its link, leaves the other end-systems unaffected. However, the central device must not fail.

In the bus topology most of the end-systems might be user workstations and the others are servers. However, in the star topology, the end-systems might be user workstations or servers but the central device is different. It is a specialised device with the purpose of connecting other devices in the network. Currently, the star topology is the usual way to configure a network. There are several reasons for this. The most important is that the central device can be used to connect the network to other networks and, in particular, to the Internet.

Discussion Point:

Which network topologies have you used? You might wish to defer this discussion until you have read about network devices later in this chapter.

In a situation where several LANs are connected, they can have different topologies or supporting technologies. This collection of LANs then becomes a **hybrid network**. A special connecting device is needed to ensure that the hybrid network is fully functional. It is often an advantage to be able to connect a new topology LAN to existing LANs where it is not sensible or not possible to use the existing topology for the new LAN. An example is when a wired LAN is already installed but a new wireless LAN is to be connected to it.

KEY TERMS

Hybrid network: a collection of connected LANs where some of them have different topologies or supporting technologies

Cable: a transmission using copper wire or fibre-optic

Bandwidth: a measure of the amount of data that can be transmitted per second

2.03 Transmission media

Cable

A network **cable** can be twisted pair, coaxial or fibre-optic. The twisted pair and coaxial cables both use copper for the transmission medium. In discussing suitability for a given application there are a number of factors to consider. One of these is the cost of the cable and connecting devices. Another is the best **bandwidth** that can be achieved. The bandwidth governs the possible data transmission rate. There are then two factors that can cause poor performance: the likelihood of interference affecting transmitted signals and the extent of attenuation (deterioration of the signal) when high frequencies are transmitted. These factors will dictate whether repeaters or amplifiers are needed in transmission lines and how many will be needed. Table 2.01 shows some comparisons of the different cable types.

	Twisted pair	Coaxial	Fibre-optic
Cost	Lowest	Higher	Highest
Bandwidth or data rate	Lowest	Higher	Much higher
Attenuation at high frequency	Affected	Most affected	Least affected
Interference	Worst affected	Less affected	Least affected
Need for repeaters	More often	More often	Less often

Table 2.01 Comparisons between cable types

You need to understand that for each of the three types of cabling there are defined standards for different grades of cable which must be considered when you decide which type of cable to use. Fibre-optic cable performs best but costs more than the other kinds. For a new installation the improved performance of fibre-optic cable is likely to be the factor that governs your choice. However, where copper cable is already installed the cost of replacement by fibre-optic cable may not be justified.

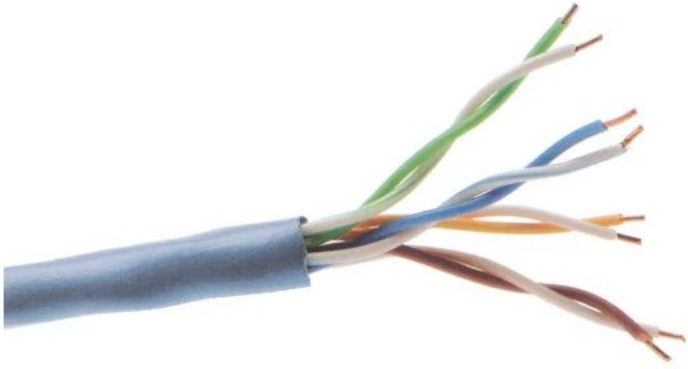


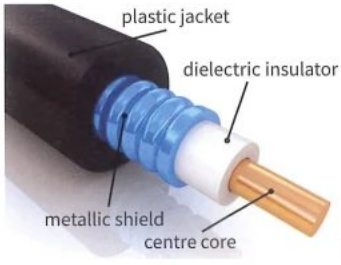
Figure 2.05 One cable with four twisted pairs with differing twist rates to reduce interference

Currently, twisted pair cable is normally used to connect telephone handsets to telephone lines. This type of cable is illustrated in Figure 2.05. It is also the technology of choice for high-speed local area networks.

Question 2.01

Twisted pair cable can be shielded or unshielded. What are the options for this? How does shielding affect the use of the cable?

Coaxial cable is used extensively by cable television companies and in metropolitan area networks. It is not usually used for long-distance telephone cabling. Fibre-optic cable is the technology of choice for long-distance cabling. As shown in Figure 2.06, coaxial cable is not bundled but a fibre-optic cable contains many individual fibres.



a



b

Figure 2.06 (a) Coaxial cable and (b) a bundled fibre-optic cable

Wireless

The alternative to cable is **wireless** transmission. The three options here are radio, microwave or infrared. These are all examples of electromagnetic radiation; the only intrinsic difference between the three types is the frequency of the waves.

KEY TERM

Wireless: a transmission using radio, microwave or infrared

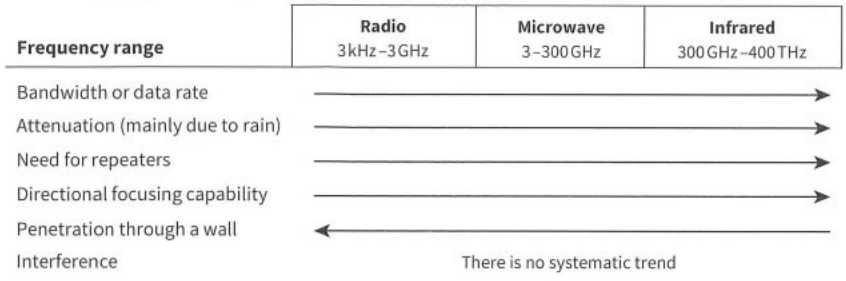


Figure 2.07 Frequency ranges and frequency dependency of factors affecting wireless transmission

When making a choice of which wireless option to use you need to consider all of the same factors that were discussed when comparing different kinds of cable. In addition, the ability of the radiation to transmit through a solid barrier is an important factor. Also, the extent to which the transmission can be focused in a specific direction needs to be considered. Figure 2.07 shows the approximate frequency ranges for the three types of radiation. The factors listed on the left increase in the direction of the arrows. The bandwidth increases through radio and microwave to infrared but the ability of the waves to penetrate solid objects is greatest for radio waves. Interference is not consistently affected by the frequency.

The increased attenuation for infrared transmission, which has the highest frequency, means that it is only suitable for indoor applications. The fact that it will not penetrate through a wall is then of benefit because the transmission cannot escape and cause unwanted interference elsewhere. For most applications, microwave transmission is the best option because it has a better bandwidth compared to that available using radio waves.

Comparing cable and wireless transmission

It is worth noting that cables are often referred to as 'guided media' and wireless as 'unguided media'. This is slightly misleading because only radio wave transmission fits the description of unguided. It is possible with microwaves or infrared to direct a transmission towards a particular receiver (as suggested in Figure 2.07).

There are other points to consider when we compare the relative advantages of transmission through a cable or wireless transmission.

- The use of certain wireless transmission frequencies is regulated by government agencies and so permission has to be obtained before wireless transmission is used.
- Outside these frequencies, no permission is needed to use the air for transmission but cables can only be laid in the ground with the permission of landowners.
- For global communications, the two competing technologies are: transmission through fibre-optic cables laid underground (or on the sea bed) and satellite transmission (discussed later in this section).
- Interference is much more significant for wireless transmission and its extent is dependent on which frequencies are being used for different applications.
- Repeaters are needed less often for wireless transmission.
- Mobile (cell) phones now dominate Internet use and for these, only wireless transmission is possible.
- For home or small office use, wired or wireless transmission is equally efficient; often, not having to install cables favours wireless connections for a small network.

Satellites are components of modern communication systems. Figure 2.08 shows the altitudes (distances above Earth) of three different types of satellite. The Van Allen belts are areas containing high levels of electrically charged particles, which interfere with satellites.

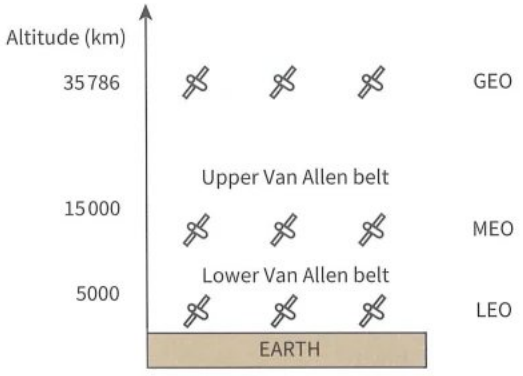


Figure 2.08 Satellite altitudes

The highest altitude satellites are in geostationary Earth orbit (GEO) over the equator and these are used to provide long-distance telephone and computer network communication. 'Geostationary' means that the satellite orbits at the same speed as the Earth spins, so from a point on the Earth the satellite always appears to be at the same point in the sky. Only three GEO satellites are needed for full global coverage. Closer to Earth are a group of medium-Earth-orbit (MEO) satellites some of which provide the global positioning system (GPS). Ten MEO satellites are needed for global coverage. Finally, low-Earth-orbit (LEO) satellites work in 'constellations' to supplement the mobile phone networks. Fifty LEO satellites are needed for full global coverage but currently there are several hundred LEO satellites in orbit.

A satellite can act as a component in a network and can directly connect with ground-based components. These ground-based components can be much further apart than in a network with no satellites. The disadvantage of satellites is that the greater transmission distance causes transmission delays, which can cause technical problems for the network.

TASK 2.01

Calculate the approximate time taken for a transmission from the surface of the Earth to a medium-Earth-orbit satellite. (Take the speed of light to be 300 000 km per second.)

The use of satellites in networks tends to be for specialised applications such as the Global Positioning System (GPS) or for Internet use in remote locations. At one stage, a lot of Internet communication was expected to make use of satellites, but the development of high-speed fibre-optic cabling at relatively low cost has reduced the need for satellites.

2.04 LAN hardware

Wired LANs

In the early years, coaxial cable was used for LANs. Nowadays, twisted pair cables are probably the most widely used networking connections, and fibre-optic cables are becoming more common. In a bus configuration the bus will consist of a series of sockets linked by cables. The ends of the bus have terminators attached that prevent signals from reflecting

back down the bus. Each end-system (which is either a user workstation or a **server**), has a short length of cable with an RJ-45 connector at each end. One end is plugged into a bus socket and the other end is plugged into the LAN port of the end-system.

In a star configuration each end-system has the same type of cable with the same connectors but the cable tends to be much longer because it has to plug into a socket on the central device.

A bus can be extended by linking two bus cables using a **repeater**. A repeater is needed because over long distances, signals become attenuated (reduced in strength), making communication unreliable. A repeater receives an input signal and generates a new full-strength signal. Sometimes a bus network is constructed in what are called segments. Two segments are connected using a **bridge**. The bridge stores the network addresses for the end-systems in the two segments it connects.

The LAN port on an end-system is connected to a **Network Interface Card (NIC)**. The NIC is manufactured with a unique network address that is used to identify the end-system in which it has been installed. The addressing system is discussed in Chapter 17 (Section 17.05). For a star network, the central device might be a hub, a **switch** or a router. The switch is by far the most likely. A switch is a connecting device that can direct a communication to a specific end-system. There is discussion of how it functions in Section 2.05. The router is discussed later in this chapter and also in Chapter 17.

Wireless LANs

WiFi (WLAN in some countries) is a term used to describe wireless Ethernet. Its formal description is IEEE 802.11. This is a wireless LAN standard that uses radio frequency transmission. The central device in a WiFi LAN is a **Wireless Access Point (WAP)**. This can be an end-system in a wired network. The WAP can communicate with an end-system in the WiFi LAN provided that the end-system has a **Wireless Network Interface Card (WNIC)** installed.



KEY TERMS

Server: a system providing a service to end-systems
Repeater: a device that connects two cables and provides a full-strength signal to the second cable
Bridge: a device that connects two segments of a LAN
Network Interface Card (NIC): a component used to identify the end-system
Switch: a connecting device that can send a unicast message
Wireless Access Point (WAP): the connecting device in a WiFi LAN
Wireless Network Interface Card (WNIC): provides the NIC function in a WiFi LAN

2.05 Ethernet

Ethernet is one of the two dominant technologies in the modern networked world. It is primarily focused on LANs. Although Ethernet was first devised in the 1970s independently of any organisation, it was later adopted for standardisation by the Institute of Electrical and Electronics Engineers (IEEE). In particular it was their 802 committee that took responsibility for the development of the protocol. The standard for a wired network is denoted as IEEE 802.3 which is sometimes used as an alternative name for Ethernet. The standard has so far evolved through five generations: standard or traditional, fast, gigabit, 10 gigabit and 100 gigabit. The gigabit part of the name indicates its data transfer speed capability.

Original (or 'legacy') Ethernet was implemented on a LAN configured either as a bus or as a star with a hub as the central device. In either topology, a transmission was broadcast type. Any message would be made available to all of the end-systems without any controlled

communication exchange between any pair of end-systems. For each message received an end-system had to check the destination address defined in the message to see if it was the intended recipient.

The use of a shared medium for message transmission has the potential for messages to be corrupted during transmission. If two end-systems were to transmit messages at the same time there would be what is described as a 'collision'. This is when the voltages associated with the transmission interfere with each other causing corruption of the individual messages. The method adopted for dealing with this was CSMA/CD (carrier sense multiple access with collision detection). This relied on the fact that if a message was being transmitted there was a voltage level on the Ethernet cable which could be detected by an end-system.

The transmitter uses the following procedure.

1. Check the voltage on the transmission medium.
2. If this indicates activity, wait a random time before checking again.
3. If no activity is detected, start transmission.
4. Continuously check for a collision.
5. If no collision is detected, continue transmission.
6. If a collision is detected, stop transmission of the message and transmit a jamming signal to warn all end-stations; after a random time, try again.

Although there might be some legacy Ethernet LANs still operating, modern Ethernet is switched. The star configuration has a switch as the central device. The switch controls transmission to specific end-systems. Each end-system is connected to the switch by a full-duplex link, so no collision is possible along that link. Because there might be high levels of activity the switch needs to be able to store an incoming message in a buffer until the cable is free for the transmission to take place. Since collisions are now impossible, CSMA/CD is no longer needed. Some further details concerning Ethernet are provided in Chapter 17 (Section 17.04).

Discussion Point:

Carry out some research about the different versions of Ethernet. Which version is implemented for the systems you use? For how long will it give sufficient performance?

2.06 The Internet infrastructure

To describe the Internet as a WAN pays little attention to its size and complexity. The Internet is the biggest internetwork in existence. Furthermore, it has never been designed as a single 'whole'; it has just evolved to reach its current form and is still evolving towards whatever future form it will take.

Internet Service Provider (ISP)

One of the consequences of the Internet not having been designed is that there is no agreed definition of its structure. However, there is a hierarchical aspect to the structure (meaning that there are several distinct 'levels' within the structure). For example, the initial function of an Internet Service Provider (ISP) was to give Internet access to an individual or company. This function is now performed by what we can call an 'access ISP'. These access ISPs then connect to what we can call 'middle tier' or regional ISPs, which in turn are connected to tier 1 (or 'backbone') ISPs. An ISP is a network and connections between ISPs are handled by Internet Exchange Points (IXPs). The tier 1 ISPs are at the top of the hierarchy, along with major Internet content providers.

Discussion Point:

How many ISPs or major Internet content providers are you familiar with?

Router

We can also think of the Internet in terms of the connections that carry the most traffic, which consist of a set of fibre-optic cables laid under the sea and across land, which can be described as a 'mesh' structure. This mesh of cables contains many points where the cables connect together, which we call nodes. At every node is a device called the **router**. Routers are found not only in the general 'mesh' of the Internet but also within the ISP networks. Each router is connected to several other routers and its function is to choose the best route for a transmission. The details of how a router works are discussed in Chapter 17 (Section 17.05).

KEY TERM

Router: a device that acts as a node on the Internet

Question 2.02

How near are you to an under-the-sea Internet fibre-optic cable?

Public switched telephone network (PSTN)

Communication systems that were not originally designed for computer networking provide significant infrastructure support for the Internet. The longest standing example is what is often referred to as POTS (plain old telephone service) but is more formally described as a PSTN (public switched telephone network). There is some discussion about how PSTNs provide that support in Chapter 17. During the early years of networking the telephone network carried analogue voice data. However, digital data could be transmitted provided that a modem was used to convert the digital data to analogue signals. Another modem was used to reverse the process at the receiving end. Such so-called 'dial-up' connections provided modest-speed, shared access when required. However, an organisation could instead pay for a leased line service that provided a dedicated, permanently connected link with guaranteed transmission speed. Typically, organisations made use of leased lines to establish WANs (or possibly MANs (metropolitan area networks)).

More recently, the PSTNs have upgraded their main communication lines to fibre-optic cable employing digital technology. This has allowed them to offer improved leased line services to ISPs but has also given them the opportunity to provide their own ISP services. In this role they provide two types of service. The first is a broadband network connection for traditional network access. The second is WiFi hotspot technology, where an access point as described in Section 2.04 has a connection to a wired network providing Internet access.

Cell phone network

For users of devices with mobile (cell) phone capability there is an alternative method for gaining Internet access. This is provided by mobile phone companies acting as ISPs. The mobile phone, equipped with the appropriate software, communicates with a standard cell tower to access the wireless telephone network, which in turn provides a connection to the Internet.

2.07 Applications that make use of the Internet

The World Wide Web (WWW)

It is common practice to talk about 'using the web' or 'using the Internet' as though these were just two different ways of saying the same thing. This is not true. The Internet is, as has been described above, an Internetwork. By contrast, the World Wide Web (WWW) is a distributed application which is available on the Internet.